



Bank runs via social networks

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Abstract

This paper constructs an agent-based model of a bank run by combining elements from two distinct agent-based modeling frameworks: fractional reserve banking and social networks. A single bank takes deposits from and makes loans to a large number of households. Households form social networks and choose to withdraw their deposits based on the actions of other households within their social networks via threshold modeling mechanisms. A liquidity event occurs if the bank has insufficient funds to satisfy all withdrawal requests. Simulation results suggest that the factors that most impact liquidity events are the type of threshold modeling mechanism employed, the degree of social reach, the incidence of social shifting, and the size of loans made.

Keywords Agent-based modeling · Banking · Bank run · Social networks

JEL Classification C00 · C69 · C63 · D00 · D91 · E00 · G00 · G21

1 Introduction

In early March of 2023, Silicon Valley Bank (SVB), a depository institution that also offered specialized services to start-up firms in the San Francisco area, suffered a run on its deposit base which ultimately caused its failure. A spillover effect two days after this incident included a bank run on and subsequent failure of Signature Bank. At the time, these were the second and third largest bank failures in U.S. history. These events were very surprising and have created renewed interest in the study of bank runs, which is important because a deeper understanding of the

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financial system allows for increased resiliency and stability of that system and the economy as a whole.

Agent-based models are particularly well-suited to model complex dynamic systems such as financial markets and banking systems.¹ This paper presents an agent-based model of a bank run that essentially combines elements of an agent-based model of fractional reserve banking along with the modeling tools of the social networks literature. The goal of the paper is to determine which aspects of the model are most important in impacting the incidence of bank failure in the event of a bank run.

The topic of social networks incorporates a rather large literature, but a major branch of that literature pertains to so-called threshold models, which involve social circles, agent interaction, and the idea that one agent's behavior is dependent on other agents' behavior.² In particular, threshold models assume that agents incorporate the behavior of the members of their social network if the number of members of their social network exhibiting that behavior exceed some threshold value. Two particularly useful ways of modeling this phenomenon are known in the literature as "infection" and "influence" models. The infection model assumes that if any single member of an agent's social network adopts some behavior then that agent will also adopt that behavior. The influence model assumes that each agent has a unique threshold percentage of the agents in its social network that must adopt the behavior before the agent adopts that behavior. This paper incorporates both of these threshold models into the study of bank runs.

The core of the model in this paper is the model of banking presented in Hamill and Gilbert (2016) which was originally designed to study the effects of the fractional reserve system on capital adequacy requirements. The model includes a single bank, and many heterogeneous interacting households who can save with and borrow from the bank, and have social networks. Furthermore, the model utilizes ideas developed in Hamill and Gilbert (2009) of mimicing a social network via the notion of a social circle whereby a social network is modeled as a collection of households within a certain "social reach" (i.e., virtual proximity) of each household.³ Household behavior is contingent on the behavior of other households within its social network. Furthermore, the concept of social shifting is employed, which allows for dynamism in the members of any particular household's social network.⁴ Household

¹ Examples of other agent-based models focused on banking but not necessarily bank runs are Berardi (2007) and Mallett (2011).

² See Bruggeman (2013) for an introduction to the social network literature. See Granovetter (1978), Granovetter and Soong (1986), and Arthur (1989) for an introduction to the threshold model literature.

³ The notion of a social circle, or, in other words, the idea that a household's social network consists of the other households within a well-defined spatial proximity to that household, is designed to capture empirical characteristics of human personal networks, such as the fact that these networks tend to be of limited size, are heterogeneous in size, and change their composition over time. Moreover, this modeling strategy represents a relatively simple way of modeling a social network. The concept was formally developed in Hamill and Gilbert (2009) and has been widely used in the literature, examples of which are Wolf et al. (2015), Sohn (2019), and Hamill (2010).

⁴ Social shifting is designed to capture the idea that the members of an individual's social network can change over time.

heterogeneity arises from differences in household income and the degree to which household behavior is impacted by the members of its social network (i.e., differences in threshold value).

A bank run can occur when some households, referred to as “runners,” decide to withdraw their deposits from the bank, which, through the dynamics of social networks and the interacting nature of households, causes other households to withdraw their deposits. This can result in a situation in which the bank exhausts its supply of reserves, which is referred to as a “liquidity event.” The paper conducts several simulation experiments to determine how changes to various aspects of the model impacts the incidence of liquidity events.

A seminal paper in the bank run literature is Diamond and Dybvig (1983) (i.e., the Diamond-Dybvig model), who develop a bank run model with two agent-types (patient and impatient) and show that their model has two Nash equilibria, one with and one without a bank run. Those authors’ work has been studied and extended in many ways, which include adding various aspects of agent-based modeling.⁵ The model in the current paper is different in that it is not limited to including only two household-types and incorporates elements of the threshold modeling mechanism. The work probably most related to the current paper is Deng et al. (2010), who examine the interacting nature of agents in a Diamond-Dybvig model. The current paper differs in that it analyzes a more broad range of factors related to the threshold modeling mechanism, such as the type of threshold model utilized and the incidence of social shifting. Furthermore, the model in the current paper also includes different loan-types and the possibility of borrower default. The current paper’s main contribution to the literature is that it incorporates various aspects of threshold models that, to my knowledge, have yet to be considered in an agent-based framework studying bank runs.

The specific case of the SVB run and subsequent banking crisis has been examined in the literature. Choi et al. (2024) argue that the SVB run was caused, in general, by concern over SVB’s balance sheet, and, in particular, the unrealized losses in its U.S. Treasury securities portfolio. Furthermore, the authors’ point out that SVB’s deposits declined steadily in the year before it failed, which suggests this run was not necessarily a very short-term phenomena. Focusing less on bank balance sheet concerns, Kelly and Rose (2025) argue that the SVB run and subsequent crisis was more related to the affected banks’ relationship with problematic economic sectors. Cipriani et al. (2024) provide evidence that the crisis involved runs on significantly more banks than the ones that failed. Wang (2024) makes a comparison between the SVB and Lehman Brothers failures, while Metrick (2024) provides a nice explanation of the SVB run and resulting crisis.

To analyze the dynamics of the bank run model, several experiments are conducted. The first experiment examines how different types of threshold modeling alter the model’s dynamics. In particular, two types of infection models and two types of influence models are considered. The two infection models differ in how initial runners are distributed throughout the household population. In the

⁵ Relatively recent examples are Deng et al. (2010), Arifovic (2019), and dos Santos and Nakane (2021).

“One-Scattered” model, initial runners are randomly scattered throughout the household population, and in the “One-Clustered” model initial runners are clustered in groups within the household population. The two influence models differ in how the threshold values specific to each household are assigned. In the “Heterogeneous-Uniform” model, household thresholds are uniformly distributed, while in the “Heterogeneous-Normal” model, household thresholds are normally distributed. The four models analyzed in the first experiment serve as benchmarks for all other experiments. The second experiment analyzes how the incorporation of social shifting affects the dynamics of the four models, while the third experiment examines how an increase in households’ social reach impacts the models. In the fourth experiment, the size of the loan made by the bank is the focus. Specifically, the four models are analyzed in a scenario in which the bank makes a relatively large-sized loan, as well as a scenario in which the bank makes a relatively small-sized loan. Lastly, experiment five looks at how the possibility of borrower default impacts the four models’ dynamics.

A bank run is inherently a phenomenon involving the behavior of individuals influencing the behavior of other individuals, so agent-based modeling is an appropriate method of analysis. Furthermore, although one of the early applications of the social networking and threshold modeling concepts was the study of new product adoption, they represent a good way to study the spread throughout a population of any type of behavior. Therefore, the bank run concept seems to be a phenomenon well-suited to analysis using these tools, which represents a significant motivation and contribution of this paper.

It is useful to speculate how the various elements of the bank run model (i.e., threshold-type, social shifting, increased social reach, loan-type, and borrower default) would, in theory, affect the incidence of liquidity events. It seems logical that any factor that increases the percentage of households who become runners would lead to an increased incidence of liquidity events. Therefore, since the threshold for a household to become a runner is lower in the infection models than in the influence models, the incidence of liquidity events should be higher in the infection models. Furthermore, because an increased social reach increases the number of households that could potentially be in a particular household’s social network, this factor should increase the incidence of liquidity events in the the infection models. On the other hand, it is not immediately clear how the other elements of the bank run model would affect new runner creation and, in turn, the incidence of liquidity events.

Turning to the results, the simulation experiments suggest the following conclusions. First, as expected, threshold-type has a significant impact on the incidence of liquidity events, with the infection models generating the highest incidence of liquidity events. Second, social shifting increases the incidence of liquidity events in the infection models but not in the influence models. Third, an increased social reach increases the incidence of liquidity events in the infection models, which was expected, but had little impact in the influence models. Fourth, the size of loans taken by households impacts the incidence of liquidity events, with relatively large loans causing a higher incidence of liquidity events. Fifth, the potential that borrowers may default on their loans does not impact the incidence of liquidity events.

Lastly, and surprisingly, increasing the percentage of households that become runners does not necessarily increase the incidence of liquidity events, which can result when the aggregate amount of withdrawal requests is not large enough to fully exhaust the bank's level of reserves.

This paper is organized as follows. Section 2 presents the bank run model, Sect. 3 discusses the simulation results, and Sect. 4 concludes.

2 The bank run model

The model is based on the banking and threshold models presented in Hamill and Gilbert (2016) and contains many heterogeneous households and a single bank representing the banking system. The model's dynamics are relatively simple in that the bank accepts deposits from households, and from these deposits the bank makes loans to households and holds reserves with the monetary authority. All households receive an idiosyncratic monthly income, and some households are initially designated as savers who place an amount of funds on deposit with the bank. All households are eligible for a single loan from the bank and use their income to make monthly required payments to the bank. Each time period in the model represents one month.

When deciding how many loans to make each month, the bank takes into consideration the amount of capital and reserves it needs to satisfy its desired reserve and capital requirements, as well as the payments for principal and interest it receives from borrowers. Bank liabilities are its deposits, bank assets are its loans and reserves, and bank equity capital consists of capital from shareholders and retained profit,

$$\text{bank liabilities} = \text{deposits}$$

$$\text{bank assets} = \text{loans} + \text{reserves}$$

$$\text{bank equity capital} = \text{capital from shareholders} + \text{retained profit}$$

The bank's target reserve ratio is the proportion of liquid assets it holds in relation to its deposits, whereas its target capital adequacy ratio is the proportion of equity capital it holds in comparison to its risk-weighted assets. Specifically,

$$\begin{aligned} \text{reserve ratio} &= \frac{\text{liquid assets}}{\text{deposits}} \\ \text{capital adequacy ratio} &= \frac{\text{capital}}{\text{risk-weighted assets}} \end{aligned} \quad (1)$$

where the bank's liquid assets consist of its reserves, retained profit, and equity capital. Since loans are deemed riskier assets than reserves, loans are given a higher risk-weighting than reserves. The bank makes only one type (i.e., dollar amount) of loan, so the bank's amount of risk-weighted assets is calculated as

$$\text{risk-weighted assets} = \text{total amount of loans} \times \text{risk-weight of loan} \quad (2)$$

The bank pays interest on deposits, charges interest on loans, and receives interest from a monetary authority on its reserves.

Because the bank only makes one type of loan, all borrowers receive the same size loan and have the same monthly payments, although the total amount owed by each borrowing household in any time period can be different from other borrowing households based on the timing of the loan initialization. The monthly loan payment is

$$\text{monthly loan payment} = \text{size of loan} \times \frac{\text{loan annualized interest rate}}{12}$$

Households must meet an income requirement in order to take a loan. Furthermore, the model allows for a certain percentage of borrowers each month to default on their loans, which provides the bank with fewer funds from which it can make loans.

Households who receive loans immediately “purchase” something from another randomly selected household by giving the “seller” household the proceeds of the loan. This “seller” household then deposits these funds into the bank, which provides the bank with more deposits from which it can make new loans. To be clear, nothing is actually purchased from the seller households. This mechanism simply ensures that the funds loaned out by the bank will be deposited back into the banking system.

Households with deposits at the bank can withdraw their funds at any time, and the bank satisfies these withdrawal requests from its reserves. A bank run can occur when households with deposits decide to withdraw their funds to the point where the bank exhausts its reserves. This dynamic is generated in the model by employing a threshold modeling mechanism, which assumes that a certain proportion of agents, referred to as “runners,” are concerned about the financial viability of the bank and therefore desire to withdraw their deposits for fear of losing those deposits due to the bank’s lack of reserves. This fear spreads throughout the household population via social networks. Using the social circle modeling concept in Hamill and Gilbert (2009), the model environment contains virtual “space” in the sense that households are placed in a digital grid consisting of cells. Each household forms a social network based on the concept of “social reach”, which assumes that a household’s social network consists of all of the other households that occupy the virtual space within a certain radial distance (i.e., number of cells) around that household. Figure 1 shows a stylized visual depiction of the model space. In the figure, the rectangle represents the model space, which, for the sake of demonstration, only includes household A, household B, and household C. The oval surrounding household A represents the area around household A in which any other households that are located within this space will belong to household A’s social network. Furthermore, household A’s social reach is the radial distance of the oval. In this representation, household B is within household A’s social network, but household C is outside of household A’s social network.



Fig. 1 This is a stylized visual depiction of the model space. The rectangle represents the total model space. The oval surrounding household A represents that household's social reach, which is the area in which any other households that are located in this space will belong to household A's social network. Here, household B is in household A's social network, but household C is not within household A's social network. Note that the area encompassing household A's social reach in relation to the total model space is not drawn to scale

The threshold modeling concept assumes that if the number of runners within a household's social network becomes large enough (i.e., achieves a certain threshold), that household will also become a runner. Furthermore, once a household becomes a runner it will never switch back to being a non-runner. Also, each time period a certain percentage of households can engage in "social shifting" whereby they move their position in the virtual space so that the members of any household's social network can change over time.

Each month, new runners withdraw their deposits, and the bank must satisfy these withdrawal requests via its reserves. Once the bank's reserves are depleted, the bank has experienced a liquidity event and is unable to continue operating.⁶

The following is a time line of events that occur each period in the model:

1. The bank collects debts (principal and interest) on its loans.
2. The bank pays interest to savers and collects interest on its reserves.
3. The seller households from the previous time period make deposits.
4. The bank makes new loans.
5. The borrower households make purchases from the seller households
6. New runner households make withdrawals.
7. Households potentially move and create new social networks.
8. New runner households are created.

⁶ In any time period, the bank is only subject to withdrawal requests from households who became runners in the previous time period. Therefore, if the number of runners reaches a maximum value and no longer changes over time, the bank is no longer subject to withdrawal requests. So, even though there are always runners present in the model, the bank is only subject to a run as long as the number of runners is increasing over time.

Figure 2 presents a flow diagram of the model which is very similar to the flow diagram presented in Hamill and Gilbert (2016) but is modified to include the actions of households in forming social networks and deciding whether to become runners. The modified parts of the flow diagram are shaded.

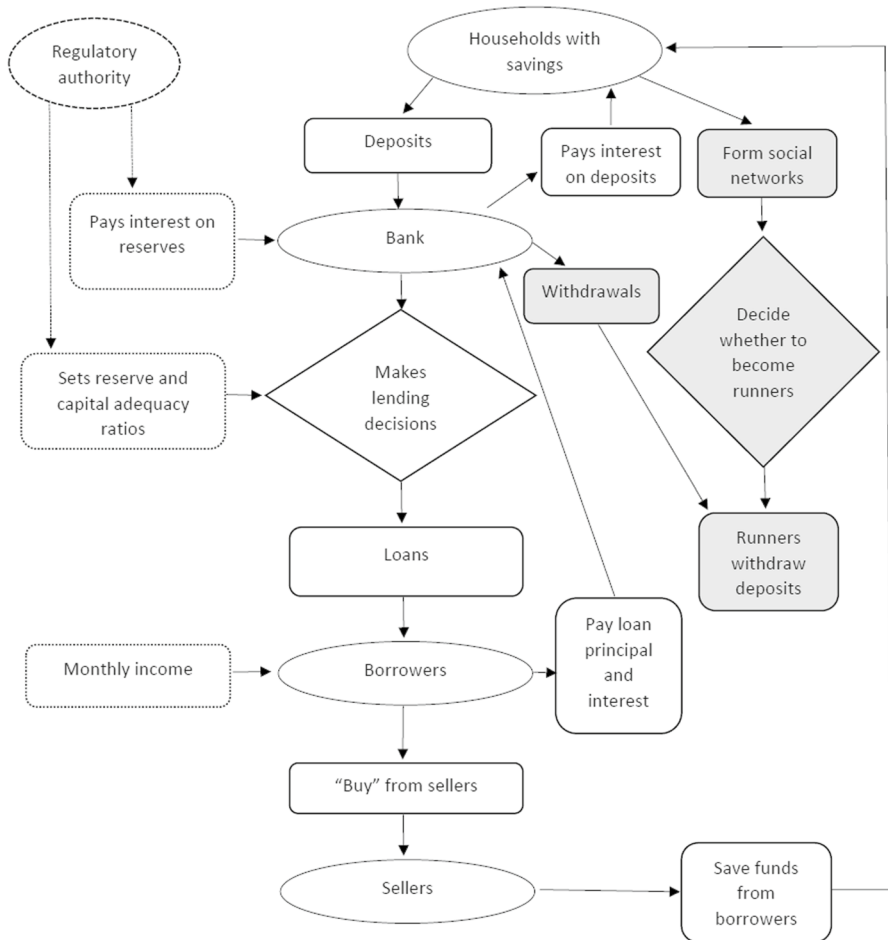


Fig. 2 This is a flow diagram of the bank run model which is very similar to the flow diagram presented in Hamill and Gilbert (2016) but is modified to include the actions of households in forming social networks and deciding whether to become runners. The shaded elements are objects that have been added to the original flow diagram in Hamill and Gilbert (2016)

3 Simulation results

In this section, several experiments are conducted to determine the aspects of the model that are most influential in impacting a liquidity event. In particular, these experiments examine the effects of threshold-type, social shifting, increased social reach, loan-type, and incidence of borrower default.

In the following experiments, the model is replicated 1000 times, with 500 time periods (i.e., months) per replication. There are 1000 households. Borrowers pay an annual interest rate of five percent on their loans, savers earn two percent annual interest on their deposits, and the bank earns two percent annual interest on its reserves. The bank's initial deposits for each replication are funded by 100 saver households who each save 10,000 dollars, which gives the bank a total of 1 million dollars of deposits. The bank has an initial amount of equity capital of 1 million dollars. The bank maintains at least a 10 percent reserve ratio and at least a 10 percent capital adequacy ratio. Each household is allocated an idiosyncratic monthly income to ensure that the average household income and minimum household income is 1000 dollars and 350 dollars, respectively. A household can only take out a loan if half of its monthly income is greater than the size of its monthly loan payment. The size of the model world is designed so that the density of households in the virtual space is 1%, so for 1000 households the two-dimensional model world is $316 \text{ cells} \times 316 \text{ cells}$, which equates to a total area of 99,856 cells, where each household occupies a single cell, and multiple households can occupy the same cell. In any replication in which a liquidity event even occurs, the simulation terminates at the time period of the liquidity event.⁷

In the following experiments the key metric is the percentage of replications in which the bank exhausts all of its reserves in the course of satisfying depositor withdrawal requests (i.e., the percentage of replications with a liquidity event). Moreover, other metrics are calculated to help in understanding that key metric. These metrics are the mean, median, and standard deviation over all of the replications of the percentage of households who have withdrawn their funds (i.e., the percentage of "runners") by the end of the replication or when the bank has experienced a liquidity event, whichever comes first. Also calculated are the mean, median, and standard deviation over the replications with a liquidity event of the time period of the liquidity event.

In any individual replication of the model, the key metrics that determine the incidence of a liquidity event are the percentage of households that are runners, as well as the bank's quantity of reserves and deposits. Therefore, to aid in understanding the dynamics of the model, for each experiment conducted, the time paths of those three metrics for a single replication of each variant of the model are presented.

⁷ Many of the values of the parameters used in these experiments are similar to those used in Hamill and Gilbert (2016).

Table 1 Threshold type

Threshold type	Liquidity event percentage	Percentage of runners			Liquidity event time period		
		Mean	Median	Std. dev	Mean	Median	Std. sev
One-Scattered	77.30	28.77	26.20	11.51	4.66	4.00	2.24
One-Clustered	8.90	20.07	19.70	8.28	4.84	3.00	4.22
Het-Uniform	4.60	7.66	7.60	1.41	2.70	3.00	0.79
Het-Normal	2.10	6.58	6.50	1.22	2.52	2.00	0.68

“One-Scattered” and “One-Clustered” are infection models in which innovators are randomly distributed and clustered throughout the model space, respectively. “Het-Uniform” and “Het-Normal” are influence models in which household thresholds are uniformly and normally distributed, respectively. The “Liquidity Event Time Period” values are calculated for replications in which a liquidity event occurred. In all of these simulations there is no social shifting, social reach is equal to ten cells, borrowing households take out loans of one-hundred thousand dollars, and no households default on their loans

3.1 Threshold-type

The first experiment examines how different variants of the threshold modeling mechanism affect the incidence of liquidity events. In the simulations in this section there are two types of threshold models utilized, an infection model and an influence model. In the infection model, if one member of a household’s social network is a “runner” (i.e., is a household that chooses to withdraw their deposits from the bank), then that household will also become a runner. In the influence model, each household has a different “threshold” by which when the percentage of runners in that household’s social network exceeds the threshold, then that household will also become a runner.

In both types of threshold models, there must be a way to determine the initial runners, which are referred to in the literature as “innovators”. Following Hamill and Gilbert (2016) and Rogers (2003), in both model-types, innovators are set to be 2.5 percent of the household population. In the infection model, innovators are either scattered throughout the household population (i.e., the “One-Scattered” model), or clustered in groups within the household population (i.e., the “One-Clustered” model). In the influence model, household thresholds are either uniformly distributed between 0 percent and 100 percent (i.e., the “Heterogeneous-Uniform” model), or normally distributed with a mean and standard deviation equal to 50 percent and adjusted to ensure that values are between 0 percent and 100 percent (i.e., the “Heterogeneous-Normal” model).⁸ In both versions of the influence model, innovators are the 2.5 percent of households with the lowest thresholds.⁹

The threshold-type employed represents an assumption about the degree of influence households have on one another. In the infection models, households have a

⁸ In the “Heterogeneous-Normal” model, any initial household threshold less than zero or greater than 100 is set to the arithmetic mean of all of the household threshold values.

⁹ The terms “One-Scattered”, “One-Clustered”, “Heterogeneous-Uniform”, and “Heterogeneous-Normal” are used in Hamill and Gilbert (2016).

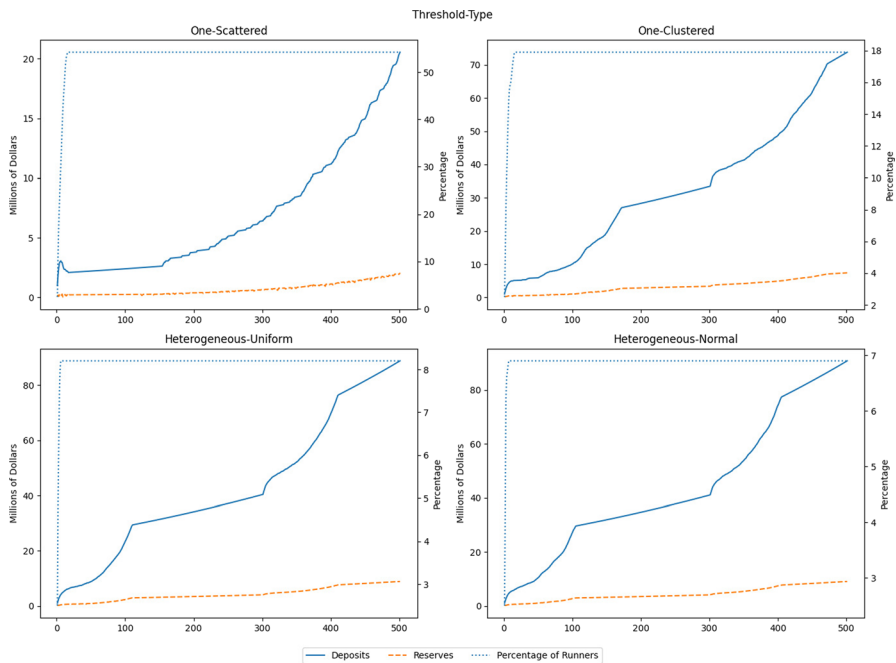


Fig. 3 These plots show the time paths of the amount of bank reserves and the amount of bank deposits (both measured on the left vertical axes), and the percentage of households who are runners (measured on the right vertical axes) for single replications of the One-Scattered, One-Clustered, Heterogeneous-Uniform, and Heterogeneous-Normal models. In these models there is no social shifting, social reach is equal to ten cells, borrowing households take out loans of one-hundred thousand dollars, and no households default on their loans

relatively high degree of influence on one-another because a household will become a runner if only a single household within its social network becomes a runner. In the influence models, households have a relatively low degree of influence on one-another because it may be necessary for a household to have more than one runner in its social network for that household to become a runner. The infection models represent a stronger assumption than the influence models.

In the following simulations, there is no social shifting, social reach is equal to ten cells, borrowing households take out loans of one-hundred thousand dollars with a three-hundred month repayment term, and no households default on their loans.¹⁰ The simulations in this section serve as benchmarks for the experiments in the later sections. Table 1 shows the results.

The infection models have higher incidences of liquidity events than the influence models. Specifically, the One-Scattered model has the highest liquidity event percentage, while the Heterogeneous-Normal model has the lowest liquidity event percentage. The One-Clustered, Heterogeneous-Uniform, and

¹⁰ A social reach of 10 cells represents a very small proportion of the total model space of 99,856 cells.

Table 2 Social shifting

Threshold type	Social shifting	Liquidity event Percentage	Percentage of runners			Liquidity event time period		
			Mean	Median	Std. dev	Mean	Median	Std. dev
One-Scattered	No	77.30	28.77	26.20	11.51	4.66	4.00	2.24
One-Scattered	Yes	100.00	34.29	25.90	20.91	20.53	5.00	39.73
One-Clustered	No	8.90	20.07	19.70	8.28	4.84	3.00	4.22
One-Clustered	Yes	97.40	68.49	75.40	22.67	187.23	184.50	115.44
Het-Uniform	No	4.60	7.66	7.60	1.41	2.70	3.00	0.79
Het-Uniform	Yes	4.60	55.18	57.40	12.62	2.72	2.50	0.89
Het-Normal	No	2.10	6.58	6.50	1.22	2.52	2.00	0.68
Het-Normal	Yes	3.20	51.18	52.50	11.31	2.47	2.00	0.72

“One-Scattered” and “One-Clustered” are infection models in which innovators are randomly distributed and clustered throughout the model space, respectively. “Het-Uniform” and “Het-Normal” are influence models in which household thresholds are uniformly and normally distributed, respectively. The “Liquidity Event Time Period” values are calculated for replications in which a liquidity event occurred. In all of these simulations social reach is equal to ten cells, borrowing households take out loans of one-hundred thousand dollars, and no households default on their loans

Heterogeneous-Normal models have much lower incidences of liquidity events than the One-Scattered model. Furthermore, there appears to be a direct relationship between the mean and median of the percentage of runners and the liquidity event percentage, which is a straightforward and intuitive result. The idea is that more runners, on average, leads to more incidences of a liquidity event. Also, the standard deviation of the percentage of runners has a direct relationship with the liquidity event percentage; a higher standard deviation of the percentage of runners corresponds with a higher liquidity event percentage. Lastly, when liquidity events occur, they occur very quickly, that is, within the first five time periods on average, in all models. The standard deviation of the time period of liquidity events is rather low for all models, as well.

Figure 3 shows the time paths of the amount of bank reserves, the amount of bank deposits, and the percentage of households who are runners for single replications of the One-Scattered, One-Clustered, Heterogeneous-Uniform, and Heterogeneous-Normal models, respectively. In all four models, the percentage of households who become runners reaches its maximum value within a few time periods, whereas the quantities of deposits and reserves grow over time. Also, the influence models appear to have very similar dynamics, which is supported by the results shown in Table 1.

3.2 Social shifting

The second experiment examines how social shifting affects the incidence of liquidity events. Social shifting captures the idea, as pointed out in Hamill and Gilbert

(2016), that in the real world, the members of an individual's social network tend to change over time. In the simulations in this section the four threshold models are each simulated in two ways, with and without social shifting, while all other aspects of the models are the same. In the models with social shifting, in each time period five percent of the household population, chosen at random, moves their position in the model space by one cell in a random direction.¹¹ In this way, the members of the social networks of all households could potentially change in each time period.¹²

In the following simulations, social reach is equal to ten cells, borrowing households take out loans of one-hundred thousand dollars with a three-hundred month repayment term, and no households default on their loans. Table 2 shows the results.

Social shifting significantly increases the liquidity event percentage in the infection models, but has little effect on this metric in the influence models. Social shifting increases the mean percentage of runners for all models, increases the median percentage of runners for all of the models except the One-Scattered model, and increases the standard deviation of the percentage of runners in all models. The result in the influence models that even though social shifting significantly increases the percentage of runners but causes almost no increase in the incidence of liquidity events is interesting and unexpected, and suggests that just increasing the number of runners in the model will not necessarily increase the instances of a liquidity event. One way this could occur is if a significant fraction of runners meets two criteria. First, the households are initial savers with ten thousand dollar deposits. Second, the households were never selected to be seller households so they do not have the relatively large one-hundred thousand dollar deposits. Having both criteria met could result in an aggregate deposit base of runners small enough that when those deposits are withdrawn, the bank is able to maintain enough reserves to satisfy those withdrawal requests.

In the infection models, social shifting significantly increases the mean and standard deviation of the time period of a liquidity event, with the effect most pronounced in the One-Clustered model. So, even though social shifting increases the instances of liquidity events in the infection models, the time it takes those events to happen, on average, increases. On the other hand, social shifting does not have a significant effect on these metrics in the influence models.¹³

Figure 4, which displays the time paths for single replications of the One-Scattered, One-Clustered, Heterogeneous-Uniform, and Heterogeneous-Normal models with and without social shifting, show how social shifting alters the dynamics of

¹¹ The model space “wraps” in the sense that if a household is located at an edge of the model space and moves in a direction that would place it out of the model space, its new location will be on the opposite side of the model space.

¹² In the real world, the members of a person's social network can change over time for a number of reasons, such as moving to a new city, changing jobs, losing touch with friends, etc. Social shifting, as applied in the bank run model, is designed to capture this idea in a simple way, and is a reasonable assumption. It should be noted that in the model, households who engage in social shifting are doing so in a randomized way.

¹³ The idea that a bank run can take significant time to materialize is supported by the fact that Silicon Valley Bank saw a decrease in its deposits over the year prior to its failure, as pointed out in Choi et al. (2024).

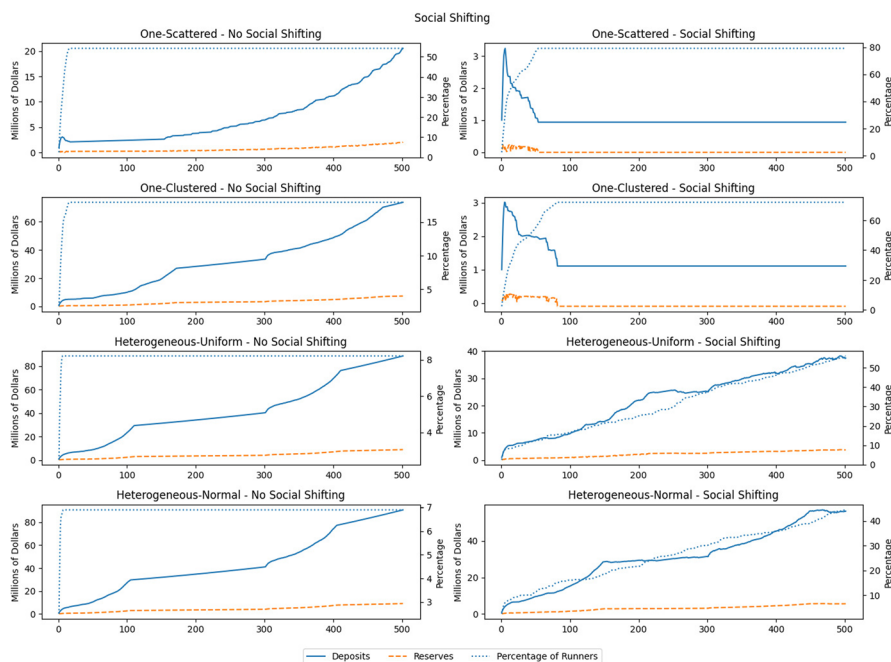


Fig. 4 These plots show the time paths of the amount of bank reserves and the amount of bank deposits (both measured on the left vertical axes), and the percentage of households who are runners (measured on the right vertical axes) for single replications of the One-Scattered, One-Clustered, Heterogeneous-Uniform, and Heterogeneous-Normal models with and without social shifting. In all of these simulations social reach is equal to ten cells, borrowing households take out loans of one-hundred thousand dollars, and no households default on their loans. Note that in the One-Scattered and One-Clustered models with social shifting, the bank experienced a liquidity event, which terminated the simulations

each of the models. Interestingly, compared to the models without social shifting, in all models the inclusion of social shifting extends the amount of time it takes for the percentage of households who become runners to achieve its maximum value. This may help explain the increase in the average time period of a liquidity event seen in the infection models in that there appears to be a direct relationship between the speed at which the number of runners hits its maximum value and the average time period of a liquidity event. However, in the influence models, no such relationship seems to exist since in those models with social shifting even after 500 periods the percentage of runners has still not reached its maximum value, but there is no increase in the average time period of a liquidity event compared to the models without social shifting.

3.3 Social reach

The third experiment examines how increased social reach affects the incidence of liquidity events. Social reach is the radial distance within which a household's social network lies. In the simulations in this section the four threshold models are each

Table 3 Increased social reach

Threshold type	Social reach	Liquidity event percentage	Percentage of runners			Liquidity event time period		
			Mean	Median	Std. dev	Mean	Median	Std. dev
One-Scattered	10	77.30	28.77	26.20	11.51	4.66	4.00	2.24
One-Scattered	30	100.00	91.26	92.25	6.08	2.98	3.00	0.12
One-Clustered	10	8.90	20.07	19.70	8.28	4.84	3.00	4.22
One-Clustered	30	100.00	35.39	37.00	12.63	3.63	4.00	0.72
Het-Uniform	10	4.60	7.66	7.60	1.41	2.70	3.00	0.79
Het-Uniform	30	1.90	9.68	8.50	5.42	2.37	2.00	0.60
Het-Normal	10	2.10	6.58	6.50	1.22	2.52	2.00	0.68
Het-Normal	30	1.20	3.07	3.00	0.49	2.00	2.00	0.00

“One-Scattered” and “One-Clustered” are infection models in which innovators are randomly distributed and clustered throughout the model space, respectively. “Het-Uniform” and “Het-Normal” are influence models in which household thresholds are uniformly and normally distributed, respectively. The “Liquidity Event Time Period” values are calculated for replications in which a liquidity event occurred. In all of these simulations there is no social shifting, borrowing households take out loans of one-hundred thousand dollars, and no households default on their loans

simulated in two ways, with a relatively short social reach of 10 cells and a relatively long social reach of 30 cells.¹⁴ All other aspects of the models are identical.

Social reach represents an assumption about how large a household’s social network can potentially be. When a household has a relatively short social reach (e.g., 10 cells), the number of households that can potentially be in that household’s social network is relatively low. When a household has a relatively long social reach (e.g., 30 cells), the number of households that can potentially be in that household’s social network is relatively high. Since, in both cases, the cell size of the social reach is very small in relation to the size of the total model space of 99,856 cells, the values seem reasonable.¹⁵

In the following simulations there is no social shifting, borrowing households take out loans of one-hundred thousand dollars with a three-hundred month repayment term, and no households default on their loans. Table 3 shows the results.

The results for the infection models are clear; an increased social reach significantly increases the liquidity event percentage, which appears to be directly related to a large increase in the mean and median percentage of runners. However, the results for the influence models conflict. While both models see a slight decrease in the incidence of liquidity events, the Heterogeneous-Uniform model shows a slight increase in the percentage of runners while the Heterogeneous-Normal model sees a slight decrease in the percentage of runners. The fact that the influence models don’t

¹⁴ Because the total model space consists of 99,856 cells, both the short social reach of 10 cells and the long social reach of 30 cells represent a very small fraction of the total model space.

¹⁵ An increased social reach can be visualized in Figure 1 by increasing the radius of the oval surrounding household A.

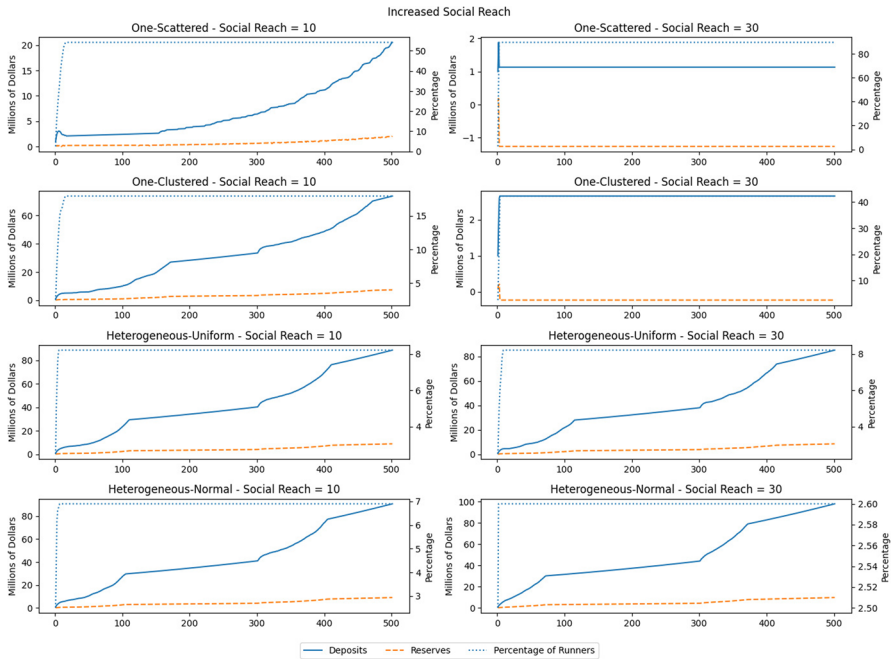


Fig. 5 These plots show the time paths of the amount of bank reserves and the amount of bank deposits (both measured on the left vertical axes), and the percentage of households who are runners (measured on the right vertical axes) for single replications of the One-Scattered, One-Clustered, Heterogeneous-Uniform, and Heterogeneous-Normal models with social reach equal to ten cells and thirty cells. In all of these simulations there is no social shifting, borrowing households take out loans of one-hundred thousand dollars, and no households default on their loans. Note that in the infection models, the bank experienced a liquidity event very quickly, which terminated the simulations

exhibit a significant increase in the percentage of runners can be explained by the notion that increasing the social reach, and thereby increasing the number of households in any particular household's social network, makes it more difficult to hit the threshold percentage of runners that's required to convert a non-runner into a runner.

With respect to the spread of the distribution of runners, the inclusion of an increased social reach in the One-Scattered and Heterogeneous-Normal models results in a decrease in the standard deviation of the percentage of runners, while the result is the opposite in the One-Clustered and Heterogeneous-Uniform models. Moreover, increased social reach seems to decrease the average time in which a liquidity event occurs in all models.

Figure 5 shows the time paths for single replications of the One-Scattered, One-Clustered, Heterogeneous-Uniform, and Heterogeneous-Normal models with social reaches equal to 10 cells and 30 cells. In all models, the percentage of households who become runners reaches its maximum value very quickly. In the infection models, the bank experienced liquidity events almost immediately, which terminated the simulations. Furthermore, an increased social reach doesn't appear to change

Table 4 Loan type

Threshold type	Loan-type	Liquidity event percentage	Percentage of runners			Liquidity event time period		
			Mean	Median	Std. dev	Mean	Median	Std. dev
One-Scattered	Mortgage loan	77.30	28.77	26.20	11.51	4.66	4.00	2.24
One-Scattered	Consumer loan	39.30	38.01	40.40	11.83	8.09	4.00	10.98
One-Clustered	Mortgage loan	8.90	20.07	19.70	8.28	4.84	3.00	4.22
One-Clustered	Consumer loan	0.00	20.75	20.00	8.24	N/A	N/A	N/A
Het-Uniform	Mortgage loan	4.60	7.66	7.60	1.41	2.70	3.00	0.79
Het-Uniform	Consumer loan	0.00	7.75	7.55	1.37	N/A	N/A	N/A
Het-Normal	Mortgage loan	2.10	6.58	6.50	1.22	2.52	2.00	0.68
Het-Normal	Consumer loan	0.00	6.47	6.40	1.14	N/A	N/A	N/A

“One-Scattered” and “One-Clustered” are infection models in which innovators are randomly distributed and clustered throughout the model space, respectively. “Het-Uniform” and “Het-Normal” are influence models in which household thresholds are uniformly and normally distributed, respectively. The “Liquidity Event Time Period” values are calculated for replications in which a liquidity event occurred. “N/A”s reflect the fact that no liquidity events occurred in that instance. In all of these simulations there is no social shifting, social reach is equal to ten cells, and no households default on their loans

the dynamics much in the influence models. These time paths are what one would expect to see based on the results in Table 3.

3.4 Loan-type

The fourth experiment examines how loan-type affects the incidence of liquidity events. In particular, each of the four models are simulated with two types of loans. One type, referred to as a “mortgage”, has a repayment term of three-hundred months and a principal amount of one-hundred thousand dollars. The second type, referred to as a “consumer loan”, has a repayment term of thirty-six months and a principal amount of five thousand dollars.¹⁶ All other aspects of the models are identical. By design, the consumer loan has a much smaller monthly payment than the mortgage loan, which will allow the experiment to determine if the size of the loan has any effect on liquidity event occurrence. In determining its capital adequacy ratio and, therefore, the number of new loans that can be made, the bank assigns mortgages and consumer loans risk weightings of 50 percent and 100 percent, respectively, which reflects the fact that consumer loans are presumably not collateralized and are therefore riskier than mortgages.¹⁷

In the following simulations there is no social shifting, social reach is equal to ten cells, and no households default on their loans. Table 4 shows the results.

The results are consistent for all models in that loan size has a similar effect on liquidity event percentage. In particular, the smaller-sized consumer loan

¹⁶ The terms “mortgage” and “consumer loan” are used to describe similar loans in Hamill and Gilbert (2016).

¹⁷ See equations (1) and (2).

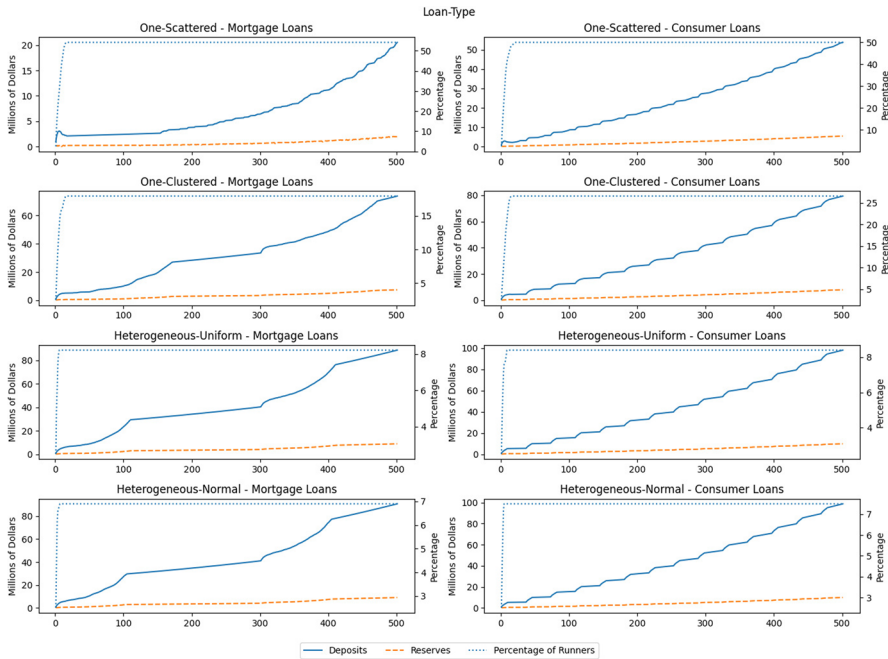


Fig. 6 These plots show the time paths of the amount of bank reserves and the amount of bank deposits (both measured on the left vertical axes), and the percentage of households who are runners (measured on the right vertical axes) for single replications of the One-Scattered, One-Clustered, Heterogeneous-Uniform, and Heterogeneous-Normal models with mortgage loans and consumer loans. In all of these simulations there is no social shifting, social reach is equal to ten cells, and no households default on their loans

significantly decreases the liquidity event percentage. In fact, consumer loans cause the complete elimination of liquidity events in the One-Clustered, Heterogeneous-Uniform, and Heterogeneous-Normal models. This result makes sense in that households are holding much smaller deposits, on average, in the consumer loan model versus the mortgage model. Therefore, when households become runners, there is a much smaller impact on the bank's deposit base than in the mortgage model where runners are holding much larger-sized loans, on average.

Consumer loans also appear to significantly increase the mean liquidity event time period for all models. Note that because there were no liquidity events in the One-clustered, Heterogeneous-Uniform, and Heterogeneous-Normal models, an “infinite” liquidity event time period can be attached to these models.

Figure 6 shows the time paths for single replications of the One-Scattered, One-Clustered, Heterogeneous-Uniform, and Heterogeneous-Normal models with mortgage loans and consumer loans. With consumer loans, the time paths of deposits appear somewhat cyclical. However, no other significant changes in the time paths are evident.

Table 5 Borrower default

Threshold type	Default	Liquidity event percentage	Percentage of runners			Liquidity event time period		
			Mean	Median	Std. dev	Mean	Median	Std. dev
One-Scattered	No	77.30	28.77	26.20	11.51	4.66	4.00	2.24
One-Scattered	Yes	76.60	28.31	25.30	11.47	4.59	4.00	2.22
One-Clustered	No	8.90	20.07	19.70	8.28	4.84	3.00	4.22
One-Clustered	Yes	8.60	19.73	19.15	7.93	4.19	3.00	3.13
Het-Uniform	No	4.60	7.66	7.60	1.41	2.70	3.00	0.79
Het-Uniform	Yes	4.20	7.66	7.60	1.37	2.67	3.00	0.69
Het-Normal	No	2.10	6.58	6.50	1.22	2.52	2.00	0.68
Het-Normal	Yes	3.60	6.46	6.40	1.17	2.33	2.00	0.68

“One-Scattered” and “One-Clustered” are infection models in which innovators are randomly distributed and clustered throughout the model space, respectively. “Het-Uniform” and “Het-Normal” are influence models in which household thresholds are uniformly and normally distributed, respectively. The “Liquidity Event Time Period” values are calculated for replications in which a liquidity event occurred. In all of these simulations there is no social shifting, social reach is equal to ten cells, and borrowing households take out loans of one-hundred thousand dollars

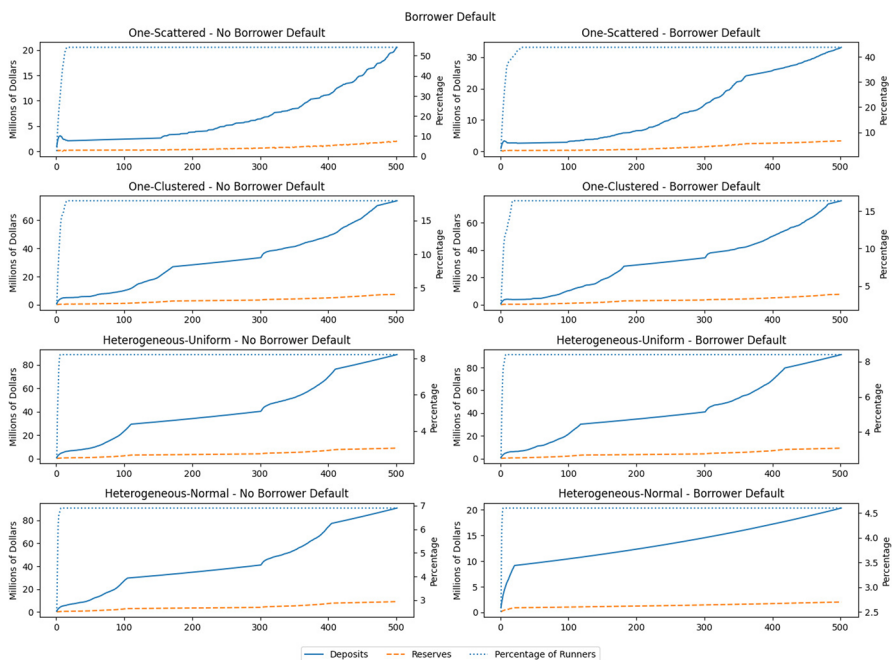


Fig. 7 These plots show the time paths of the amount of bank reserves and the amount of bank deposits (both measured on the left vertical axes), and the percentage of households who are runners (measured on the right vertical axes) for single replications of the One-Scattered, One-Clustered, Heterogeneous-Uniform, and Heterogeneous-Normal models with and without borrower default. In all of these simulations there is no social shifting, social reach is equal to ten cells, and borrowing households take out loans of one-hundred thousand dollars

3.5 Borrower default

The last experiment analyzes how the possibility of borrower default affects the incidence of liquidity events. In this section each of the four models are simulated under two scenarios. The first is one in which no borrowers default on their loans, while the second is one in which one percent of borrowers chosen at random default on their loans in the twelfth time period (i.e., month). All other aspects of the models are the same.¹⁸

In the following simulations there is no social shifting, social reach is equal to ten cells, and borrowing households take out loans of one-hundred thousand dollars with a three-hundred month term. Table 5 shows the results.

The results show that none of the models display any significant difference by including the possibility of borrower default.

Figure 7 shows the time paths for single replications of the One-Scattered, One-Clustered, Heterogeneous-Uniform, and Heterogeneous-Normal models with and without borrower default. In agreement with the results of Table 5, including the potential of borrower default does not seem to alter the dynamics of any of the models in a meaningful way.

3.6 Simulation results summary

In general, the simulation experiments suggest the following main results:

- Threshold-type has a significant impact on the incidence of liquidity events, with the infection models, and the One-Scattered model in particular, generating the highest incidence of these events.
- Social shifting increases the incidence of liquidity events in the infection models, but not in the influence models.
- Increased social reach increases the incidence of liquidity events in the infection models, but not in the influence models.
- Loan size has a significant impact on the incidence of liquidity events, with relatively large loans causing a higher incidence of these events.
- The potential of borrower default does not significantly change the incidence of liquidity events.
- An increase in the percentage of runners does not necessarily lead to an increase in the incidence of liquidity events.

¹⁸ The twelfth month was chosen arbitrarily.

4 Conclusion

This paper presented an agent-based model of a bank run in order to determine which aspects of the model are most important in impacting the incidence of bank failure in the event of a bank run. The model combined a simple agent-based model of fractional reserve banking and an agent-based threshold model of social demand that takes into account social networks and agent interaction. In general, two types of threshold modeling mechanisms were utilized, known as infection and influence models. Several simulation experiments were conducted to examine how various aspects of the model affected the incidence of the bank being unable to satisfy depositor withdrawal requests, referred to as a liquidity event. The experiments suggested six major findings. First, the type of threshold modeling mechanism utilized had a significant impact on the incidence of liquidity events, with the infection models generating the highest incidence of liquidity events. Second, the inclusion of social shifting increased the incidence of liquidity events in the infection models, but not in the influence models. Third, increased social reach increased the incidence of liquidity events in the infection models, but not in the influence models. Fourth, larger loan sizes tend to increase the incidence of liquidity events. Fifth, the potential of borrower default did not significantly change the incidence of liquidity events. Sixth, an increase in the percentage of households who become runners does not necessarily increase the incidence of liquidity events. This last finding can happen in a scenario in which the aggregate amount of withdrawal requests is not large enough to fully exhaust the bank's level of reserves.

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Author contributions C.E. conducted all work on the manuscript.

Data availability The data that support the findings of this study are available from the author upon request.

Declarations

Conflict of interest The author declares no conflict of interest.

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